Open Science: Building models like we build open-source software

Steven Kolawole

EuroPython 2022

14/07/22

Steven Kolawole

- Researcher at ML Collective, from Nigeria
- MLE at ASAlytics; opinion mining dias for algorand blockchain
- Low-budget philosopher
- Unapologetic foodie



Background

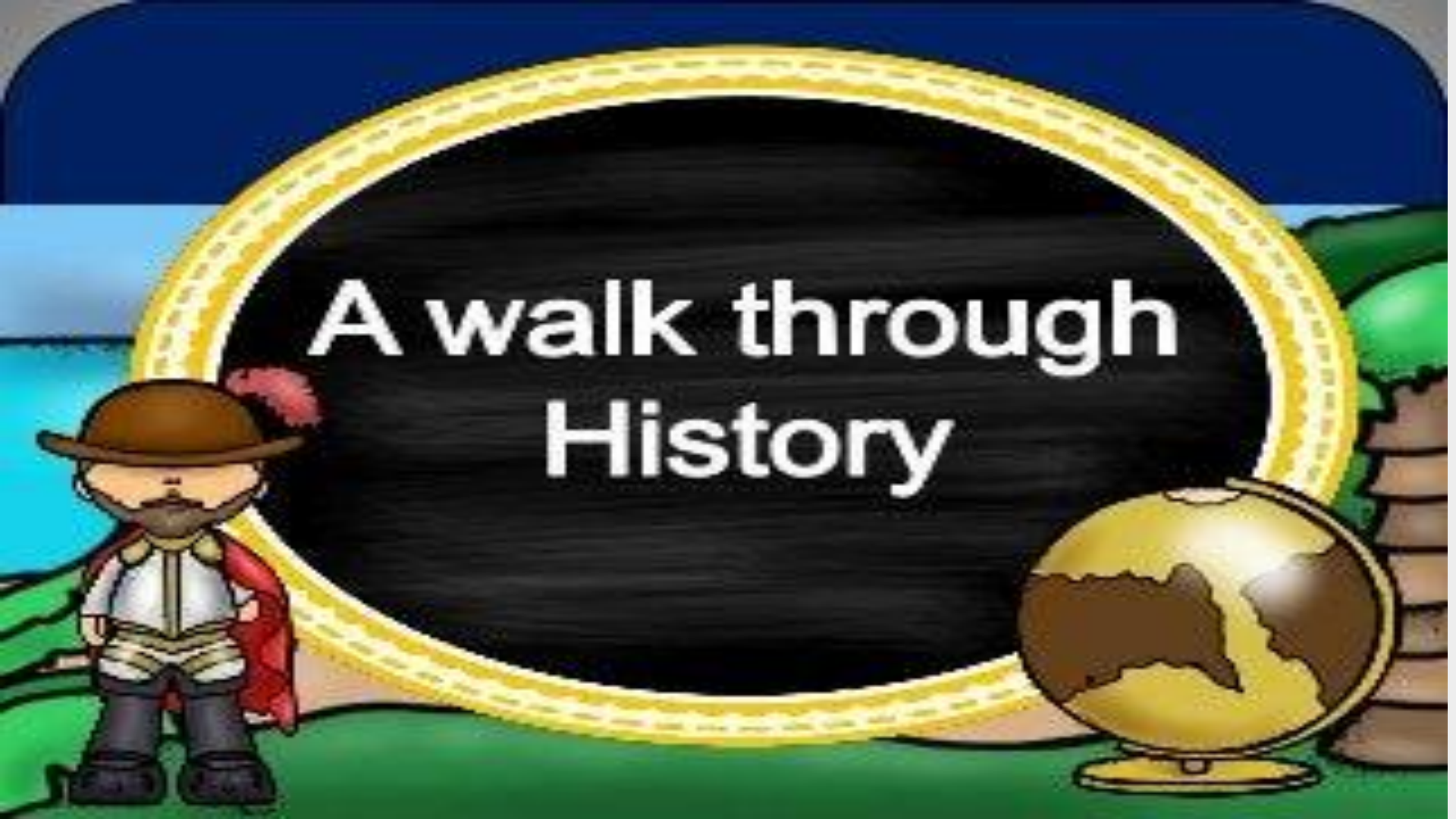
A Call to Build Models Like We Build Open-Source Software

Colin Raffel

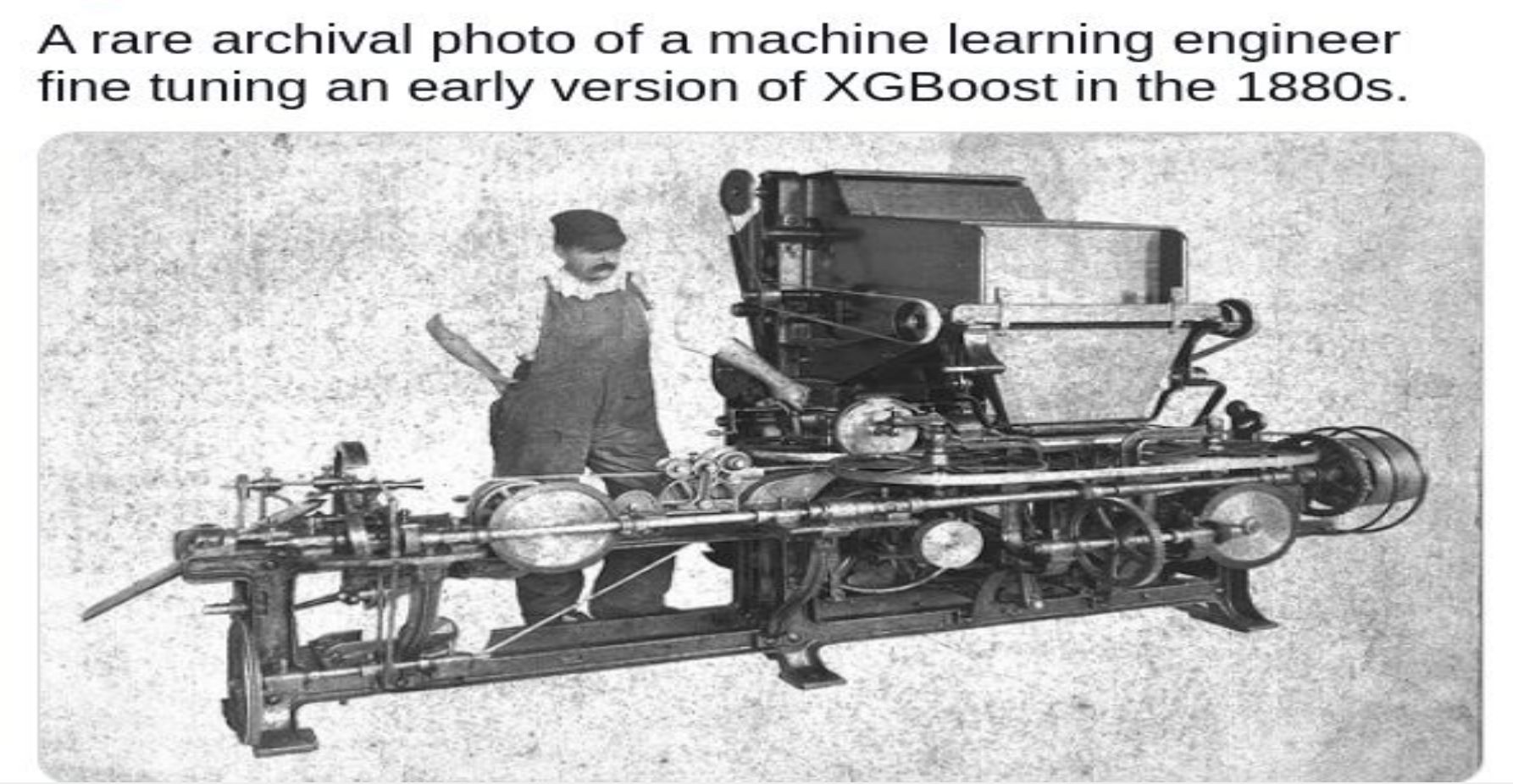
December 8th, 2021

colinraffel.com/blog

Reference:
https://colinraffel.com/blog/a-call-to-build-models-like-we-build-open-source-software.ht

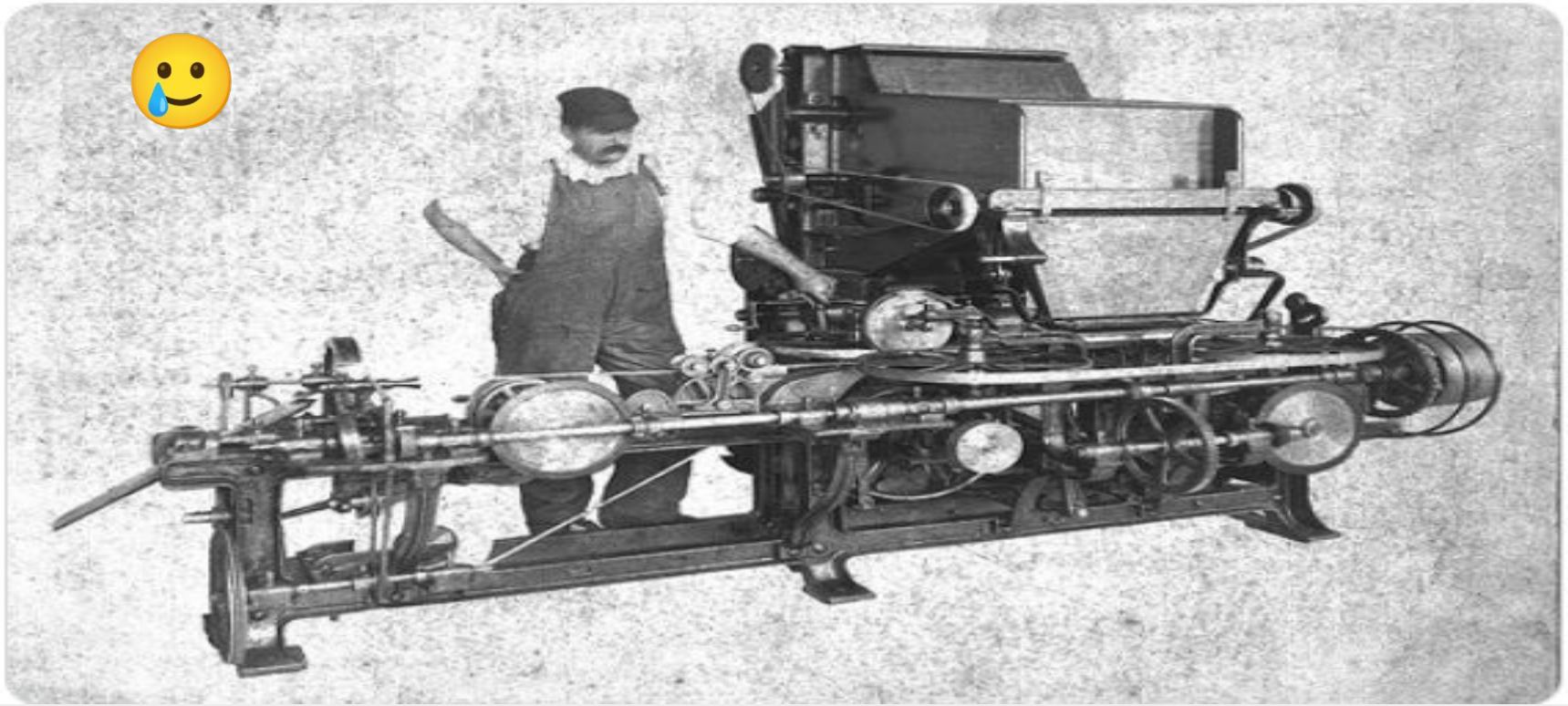

A walk through
History

Timeline: The Old World

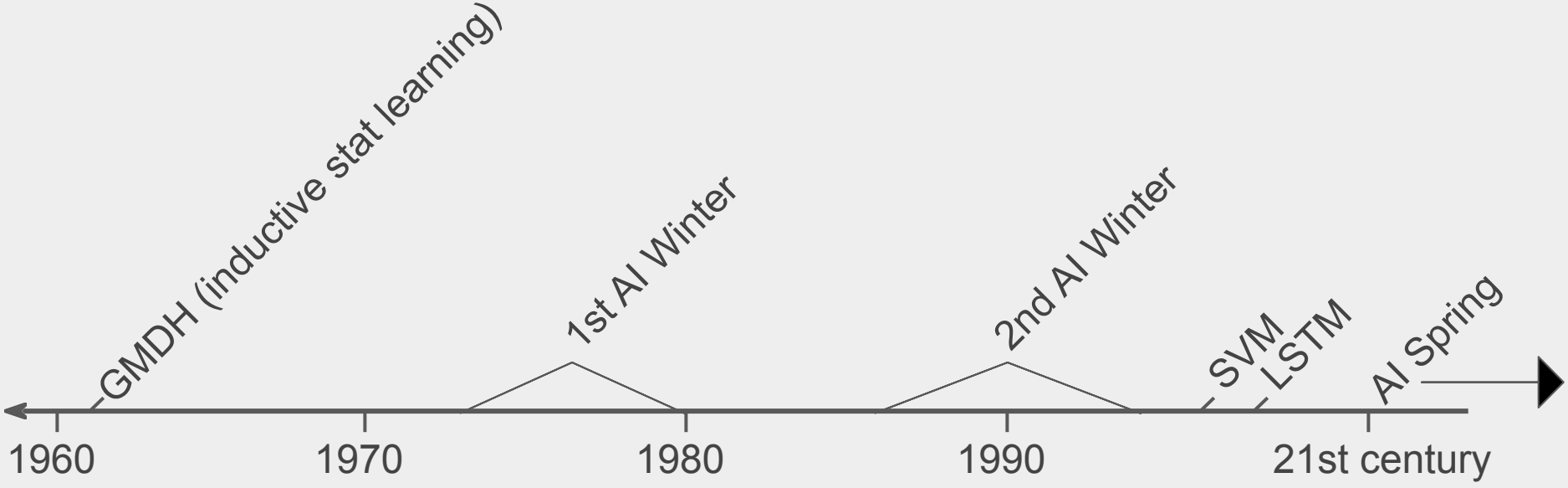


Timeline: The Old World

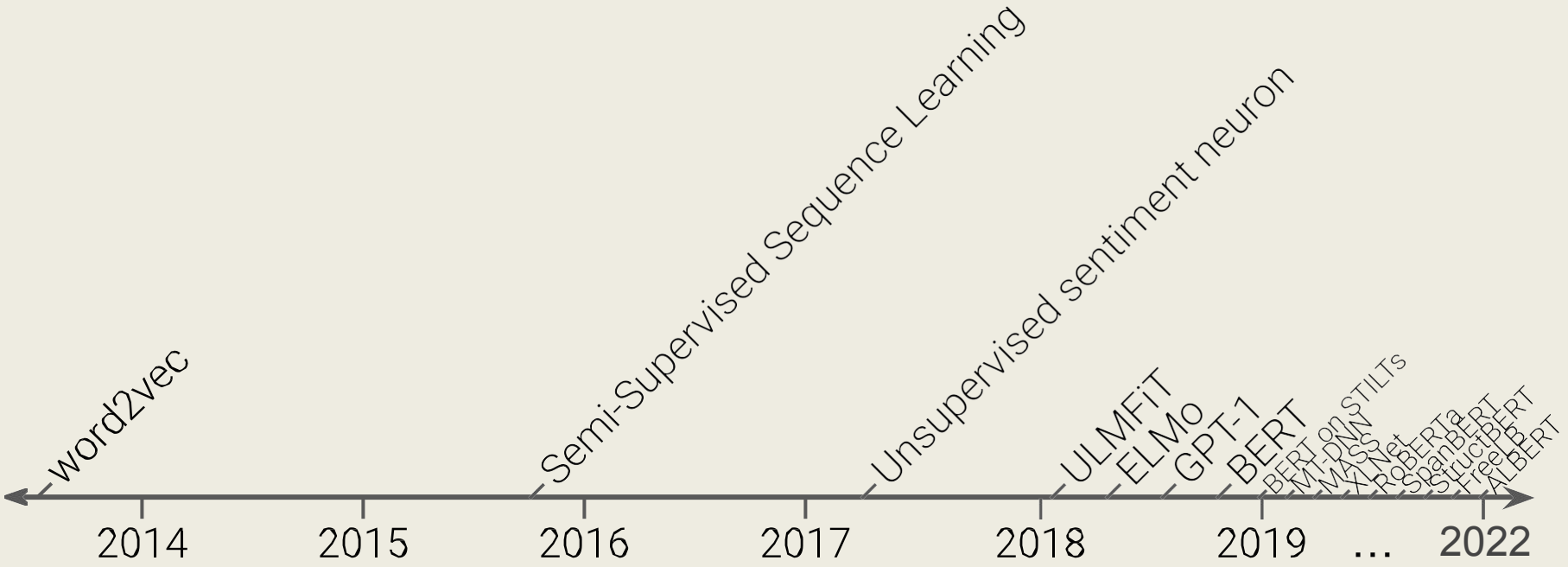
A rare archival photo of a machine learning engineer fine tuning an early version of XGBoost in the 1880s.



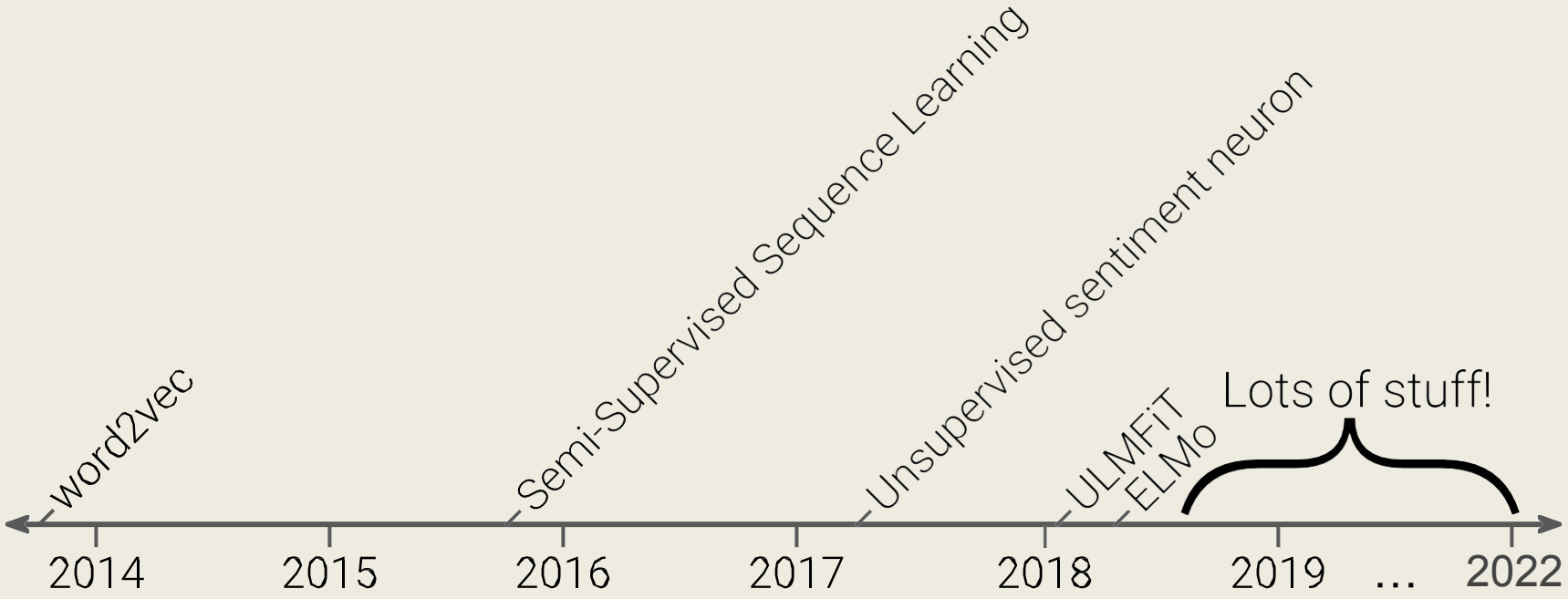
Timeline: The Old World



Timeline: Modern Era



Timeline: Modern Era



Timeline: Modern Era

- Paper A proposes an unsupervised pre-training technique called "FancyLearn".
- Paper B proposes another pre-training technique called "FancierLearn" and achieves better results.
- Paper A uses **Wikipedia** for unlabeled data.
- Paper B uses **Wikipedia and the BBC data** .

Timeline: Modern Era

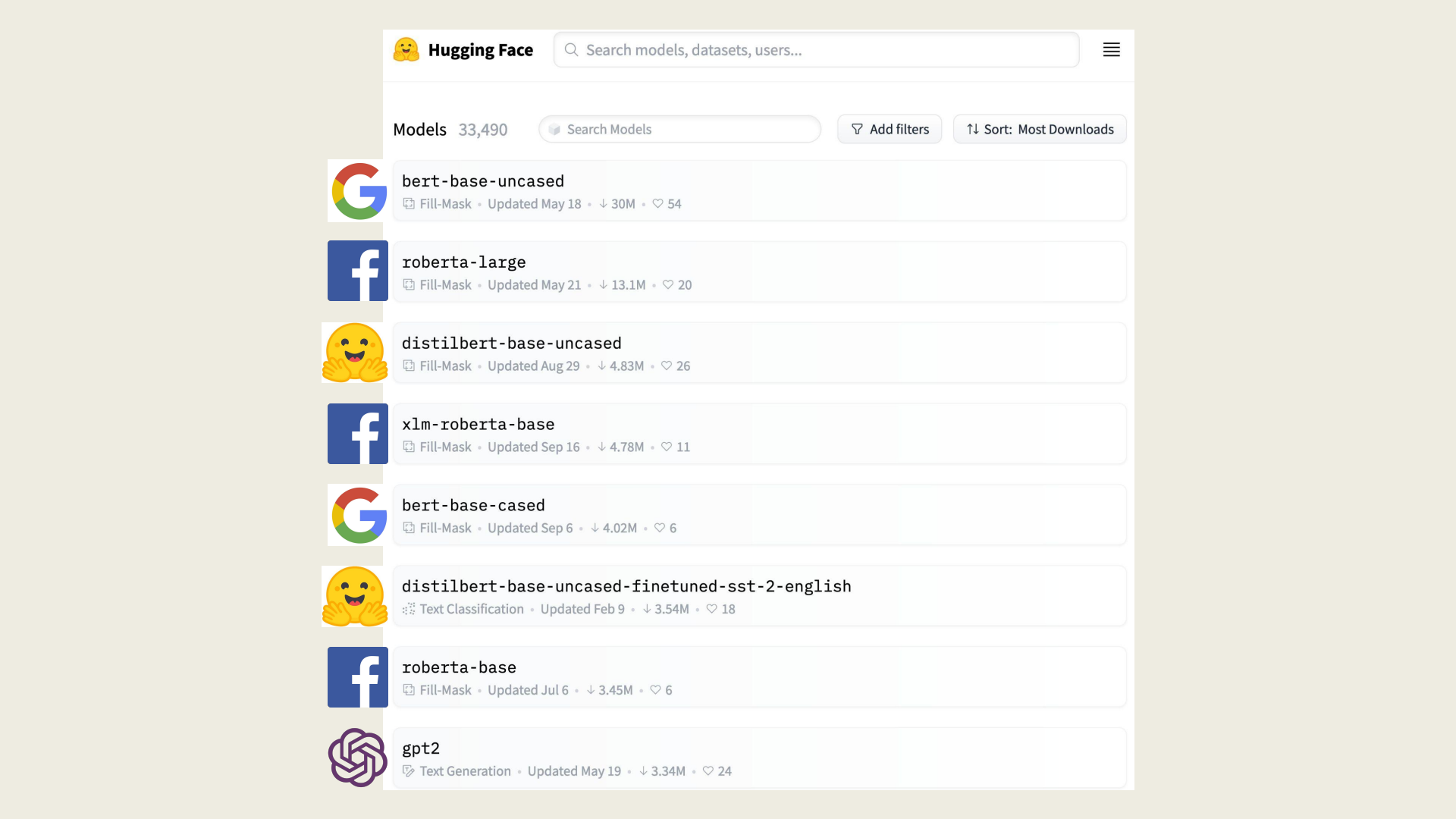
- Paper A proposes an unsupervised pre-training technique called "FancyLearn".
- Paper B proposes another pre-training technique called "FancierLearn" and achieves better results.
- Paper A uses a model with 100 million parameters.
- Paper B uses a model with 200 million parameters.

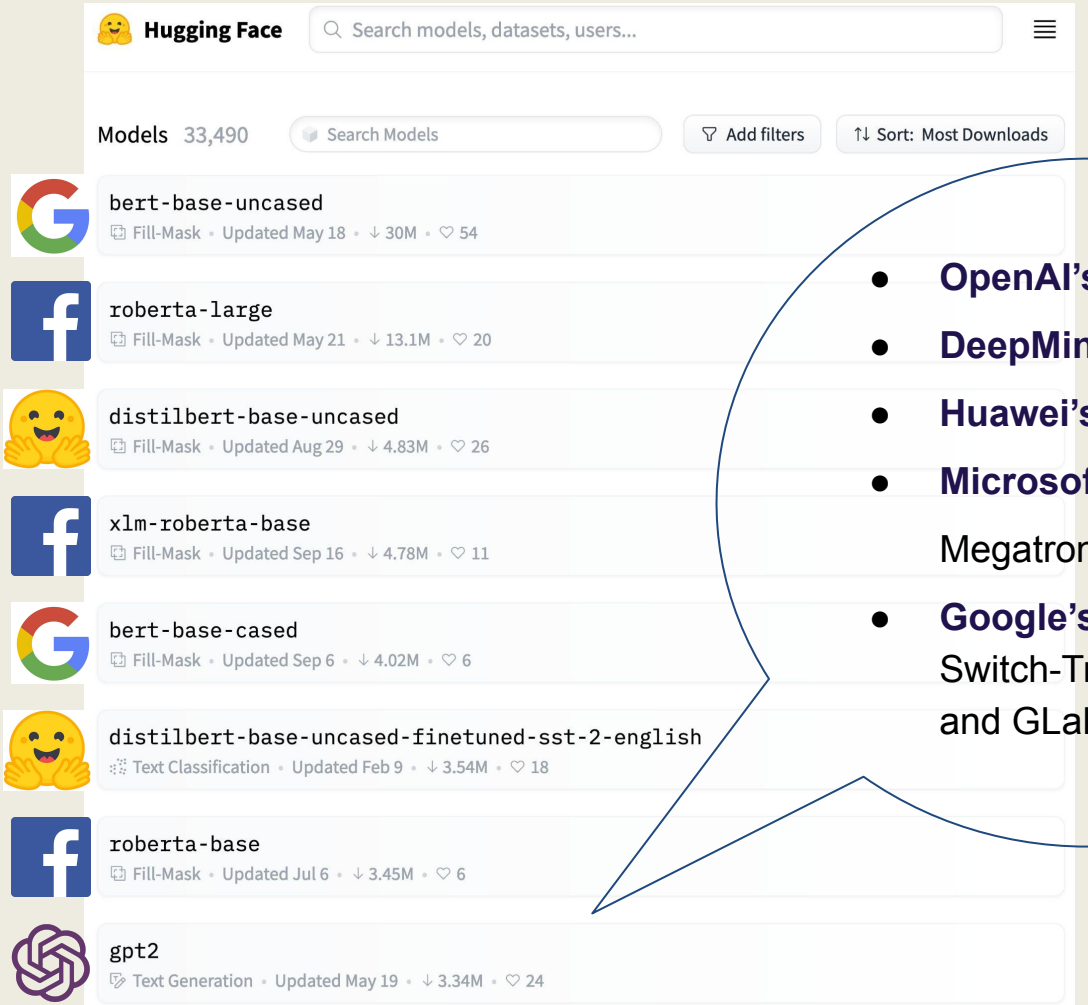
Timeline: Modern Era

- Paper A proposes an unsupervised pre-training technique called "FancyLearn".
- Paper B proposes another pre-training technique called "FancierLearn" and achieves better results.
- Paper A pre-trains on **100 billion tokens** of unlabeled data.
- Paper B pre-trains on **200 billion tokens** of unlabeled data.

Timeline: Modern Era

- Paper A proposes an unsupervised pre-training technique called "FancyLearn".
- Paper B proposes another pre-training technique called "FancierLearn" and achieves better results.
- Paper A uses the **Adam optimizer**.
- Paper B uses **SGD with momentum**.


Hugging Face
Search models, datasets, users...
Models 33,490
Search Models
Add filters
Sort: Most Downloads
bert-base-uncased
Fill-Mask • Updated May 18 • ↓ 30M • ♡ 54
roberta-large
Fill-Mask • Updated May 21 • ↓ 13.1M • ♡ 20
distilbert-base-uncased
Fill-Mask • Updated Aug 29 • ↓ 4.83M • ♡ 26
xlm-roberta-base
Fill-Mask • Updated Sep 16 • ↓ 4.78M • ♡ 11
bert-base-cased
Fill-Mask • Updated Sep 6 • ↓ 4.02M • ♡ 6
distilbert-base-uncased-finetuned-sst-2-english
Text Classification • Updated Feb 9 • ↓ 3.54M • ♡ 18
roberta-base
Fill-Mask • Updated Jul 6 • ↓ 3.45M • ♡ 6
gpt2
Text Generation • Updated May 19 • ↓ 3.34M • ♡ 24


Hugging Face
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Models 33,490
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Fill-Mask • Updated Jul 6 • ↓ 3.45M • ♡ 6
gpt2
Text Generation • Updated May 19 • ↓ 3.34M • ♡ 24
OpenAI's GPT-3
DeepMind's Gopher
Huawei's PanGu
Microsoft's
Megatron-Turing NLG
Google's
Switch-Transformer
and GLaM

*"GPT-3 175B model required 3.14E23 FLOPS of computing for training. Even at theoretical 28 TFLOPS for V100 and lowest 3 year reserved cloud pricing we could find, this will take 355 GPU-years and cost **$4.6M** for a single training run."*

Reference:
https://lambdalabs.com/blog/demystifying-gpt-3/

co:here

API

AI21 studio

Custom language models built for scale

Build sophisticated language applications on top of AI21's language models

Microsoft Megatron-Turing NLG 530B

The World's Largest and Most Powerful Generative Language Model

SambaNova SYSTEMS

PRODUCTS SOLUTIONS RESOUR

Enterprise-Grade Large Language Models Made Simple & Accessible

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OpenAI API Beta ABOUT EXAMPLES DOCS PRICING LOG IN JOIN ›

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Apply our API to any language task — semantic search, summarization, sentiment analysis, content generation, translation, and more — with only a few examples or by specifying your task in English.

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Introducing the LightOn Muse API

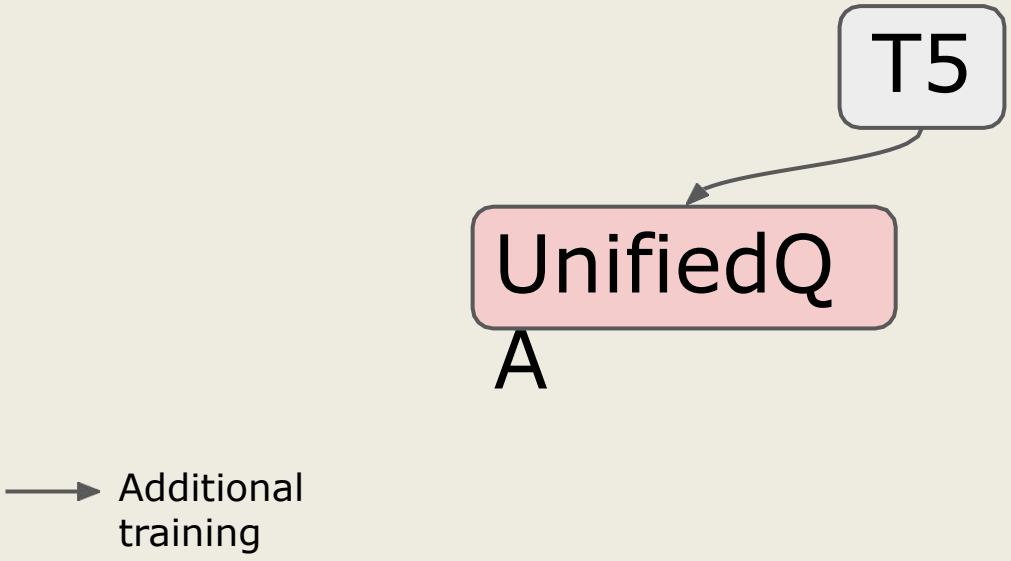
Create. Process. Understand. Learn.

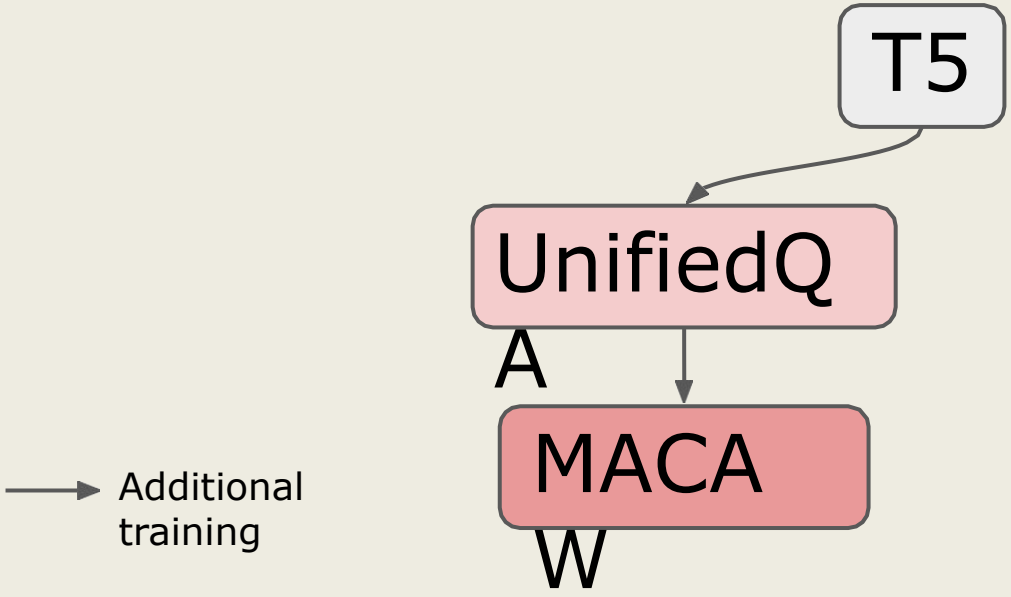
Production-ready intelligence primitives powered by state-of-the-art language models.
For the first time natively in French, Spanish, Italian, and more. **Now in private beta!**

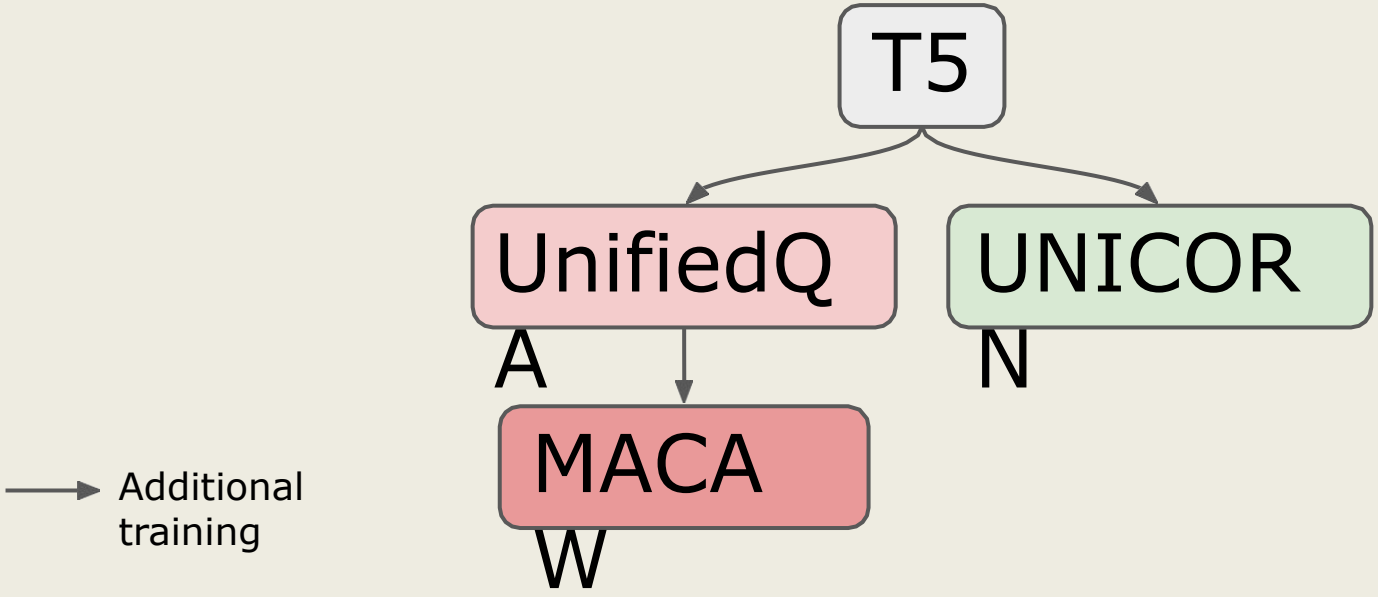
AI, 모두의 능력이 되다. HyperCLOVA

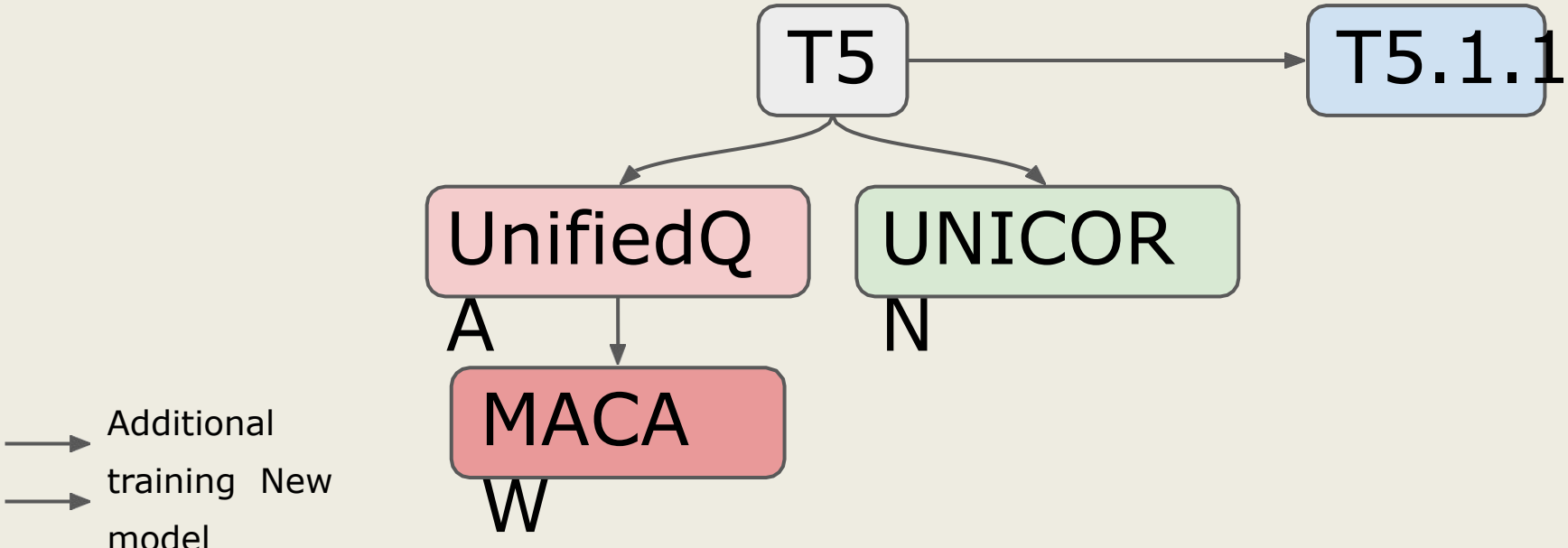
AI가 모두의 능력이 되는 새로운 시대.
그 시작이 될 HyperCLOVA를 소개합니다.
네이버 클로바와 함께 새로운 시대를 시작하세요.

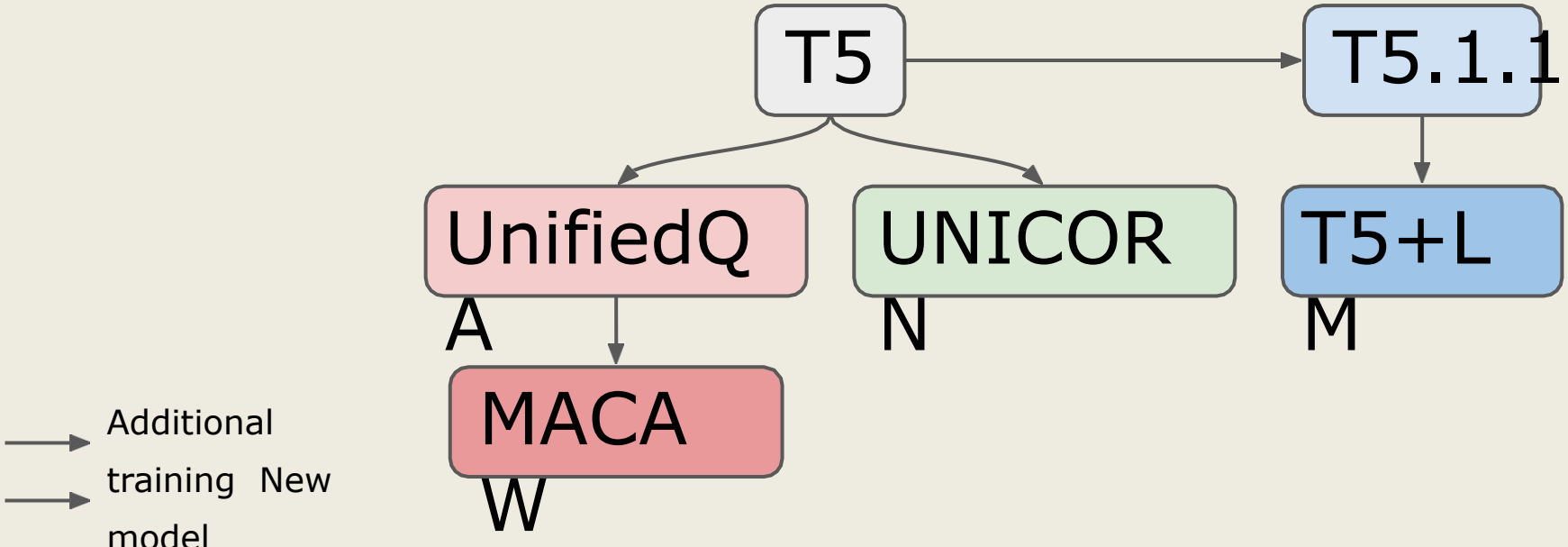
T5

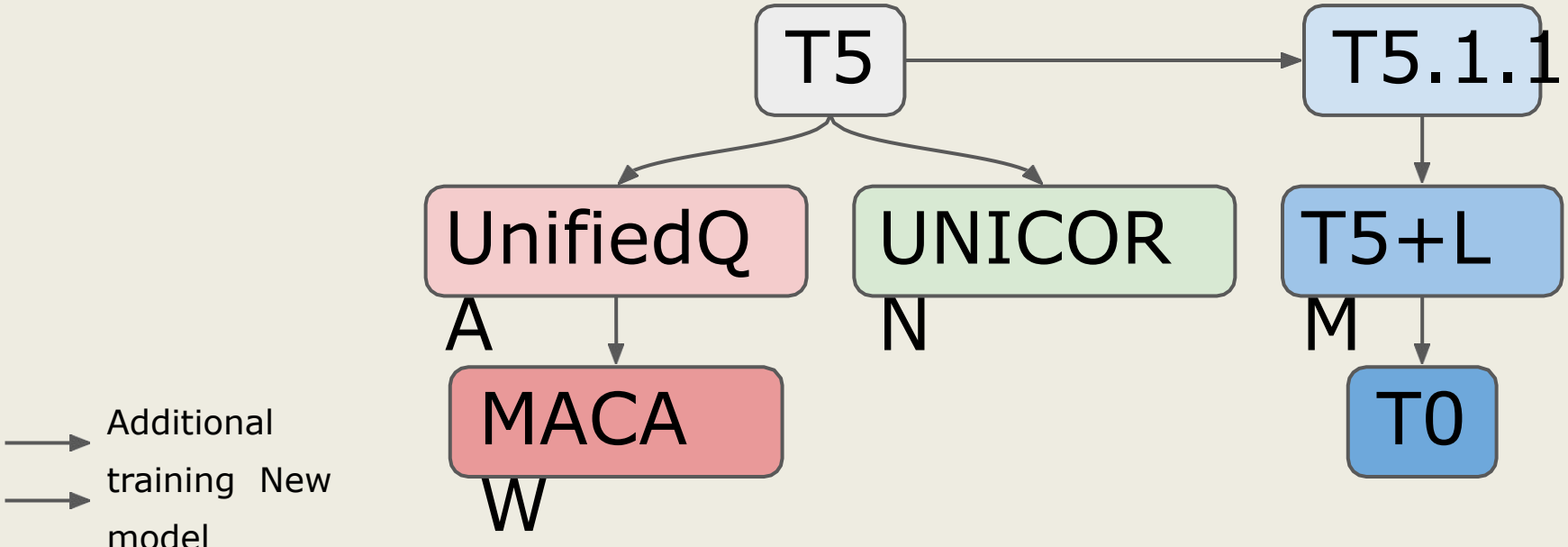

T5
UnifiedQA
Additional training

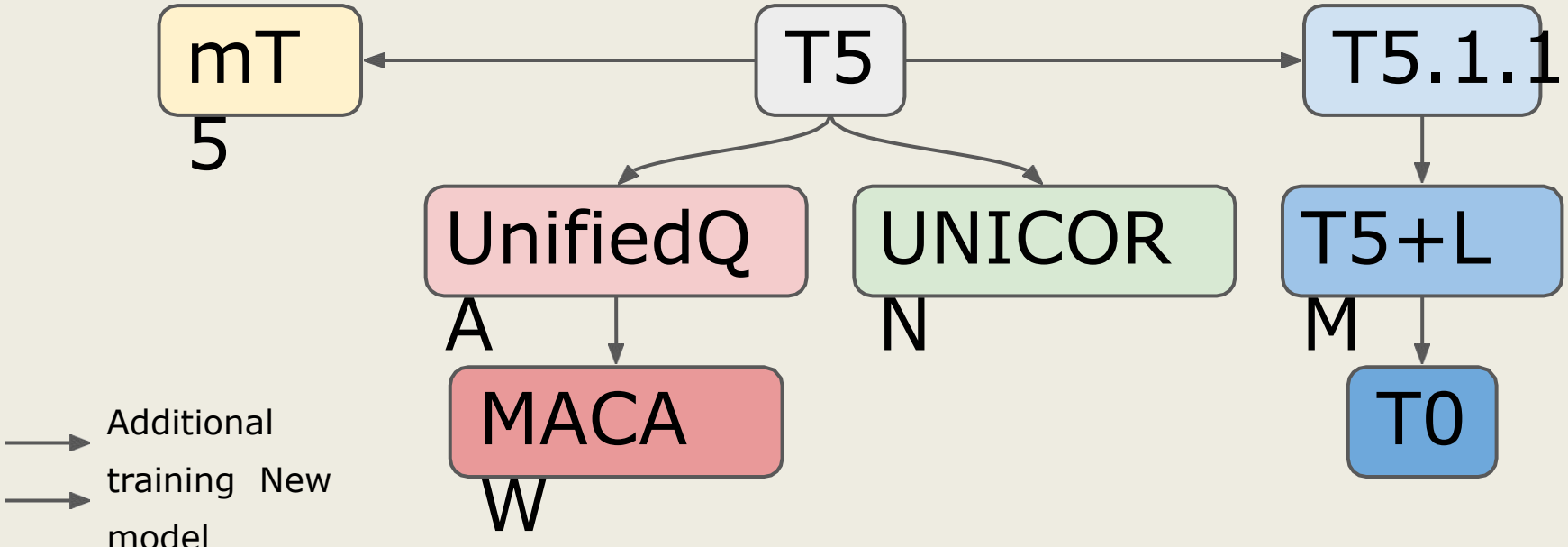

T5
UnifiedQ
A
MACA
W
Additional training

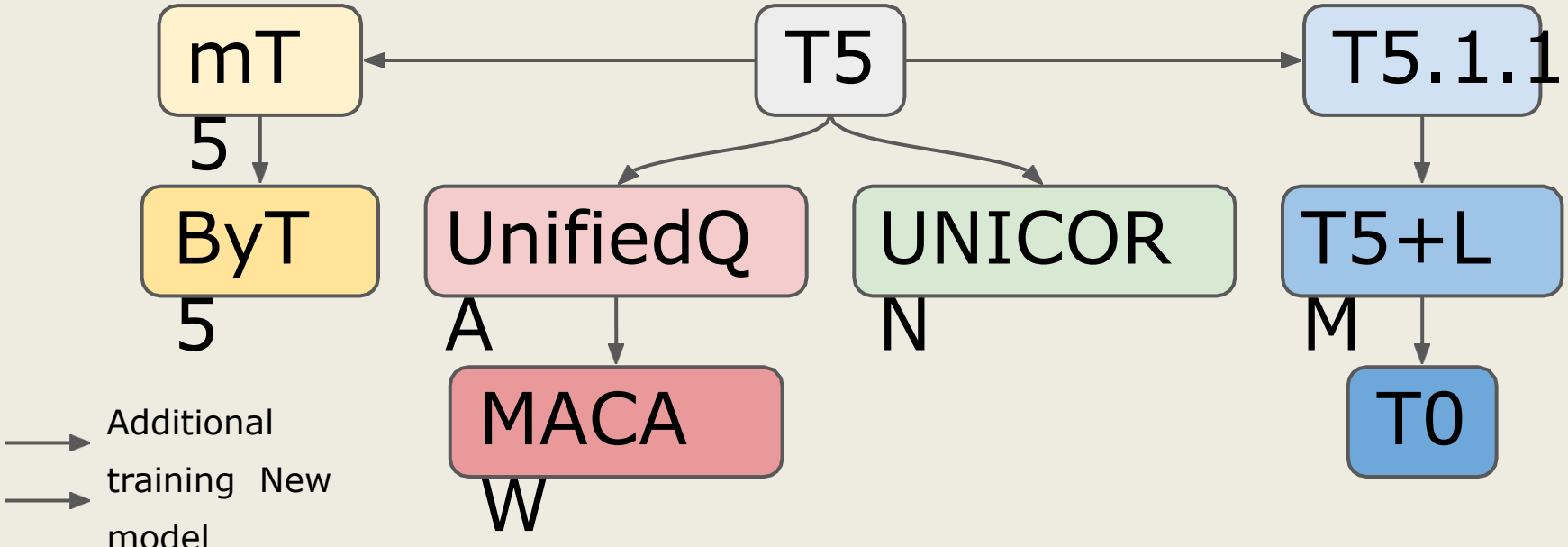

T5
UnifiedQA
UNICORN
MACAW
Additional training

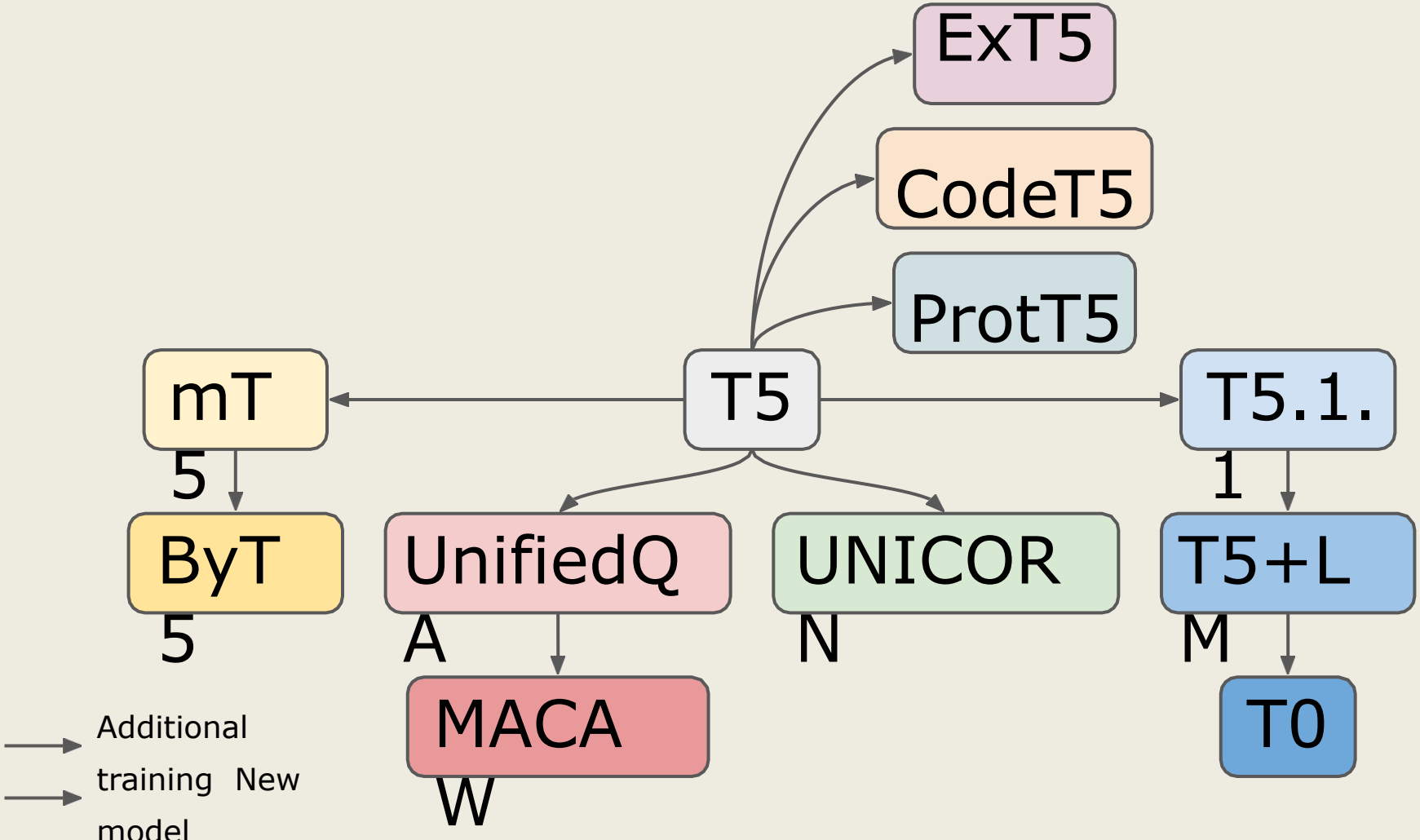

T5
T5.1.1
UnifiedQA
UNICORN
MACAW
Additional training
New model


T5
T5.1.1
UnifiedQ
A
UNICOR
N
T5+L
M
MACA
W
Additional training
New model

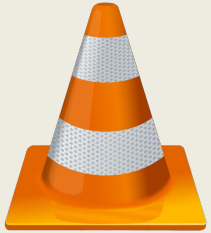
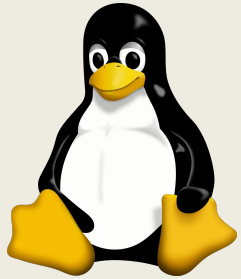

T5
T5.1.1
UnifiedQA
UNICORN
T5+LM
MACAW
T0
Additional training
New model


mT5
T5
T5.1.1
UnifiedQA
UNICORN
T5+LM
MACAW
T0
Additional training
New model


mT5
T5
T5.1.1
ByT5
UnifiedQA
UNICORN
T5+LM
MACAW
T0
Additional training
New model


ExT5
CodeT5
ProtT5
mT5
T5
T5.1.1
ByT5
UnifiedQA
UNICORN
T5+LM
MACAW
T0
Additional training
New model











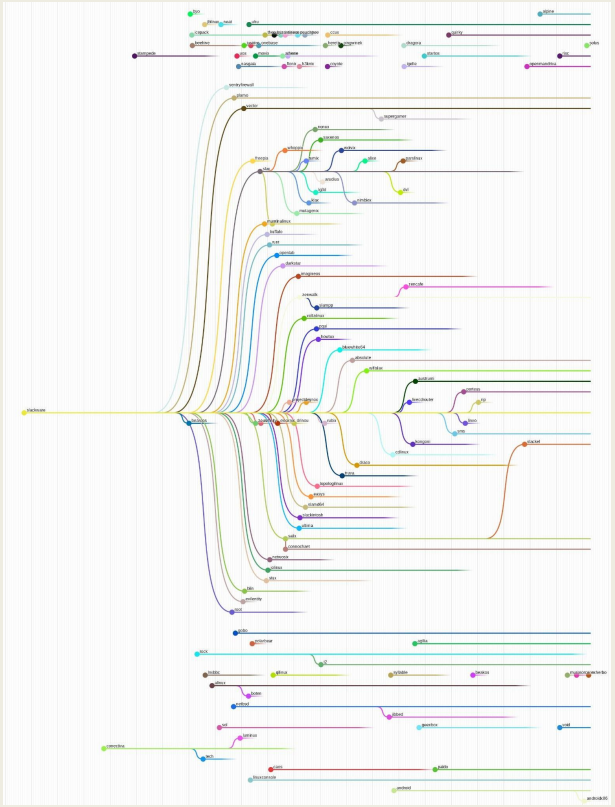


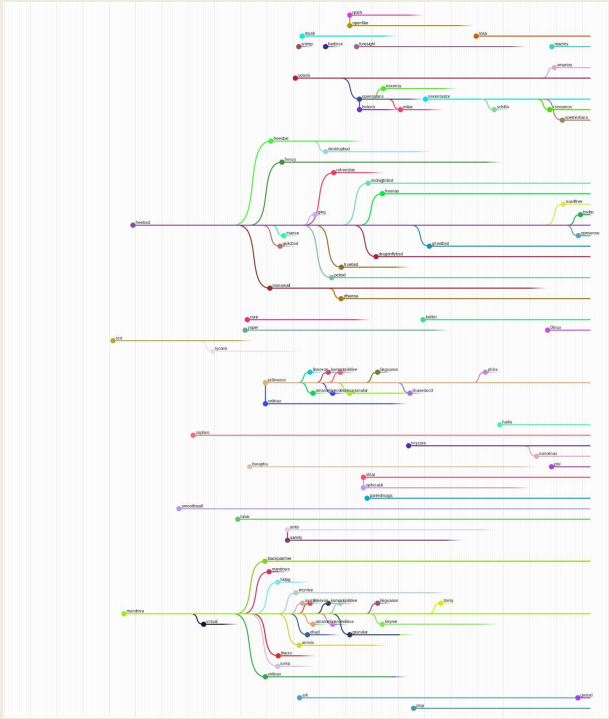






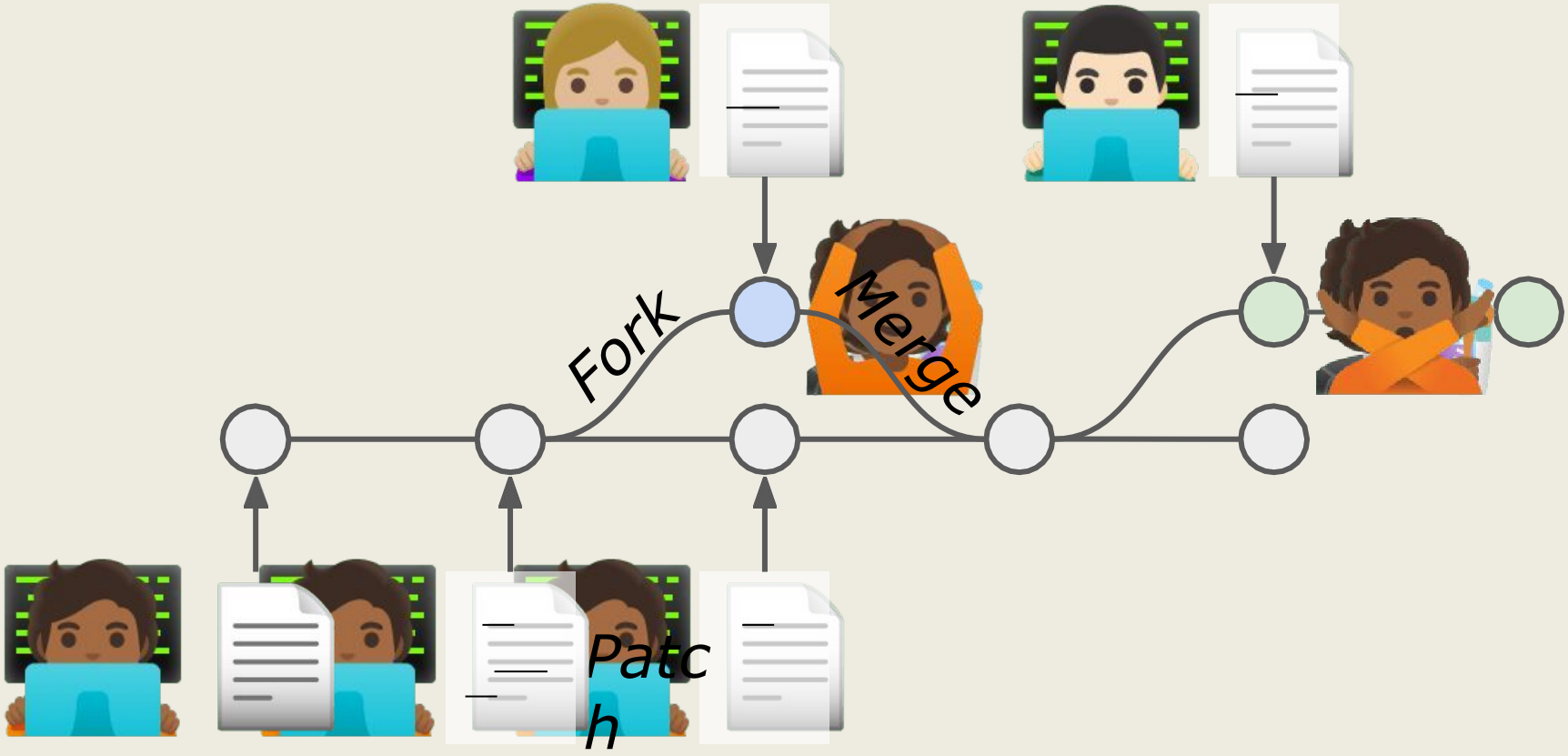

Lineage tree of Linux Distributions

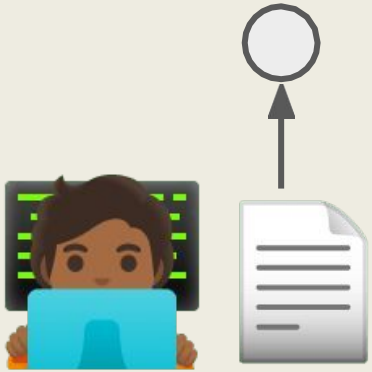


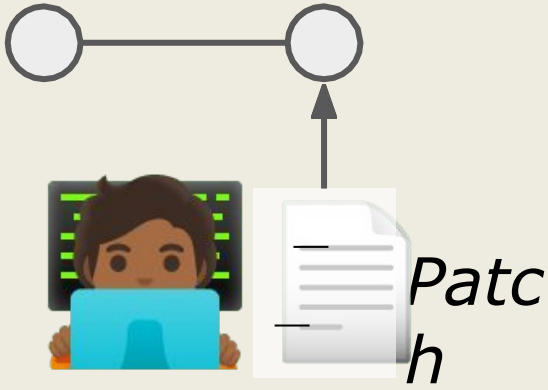


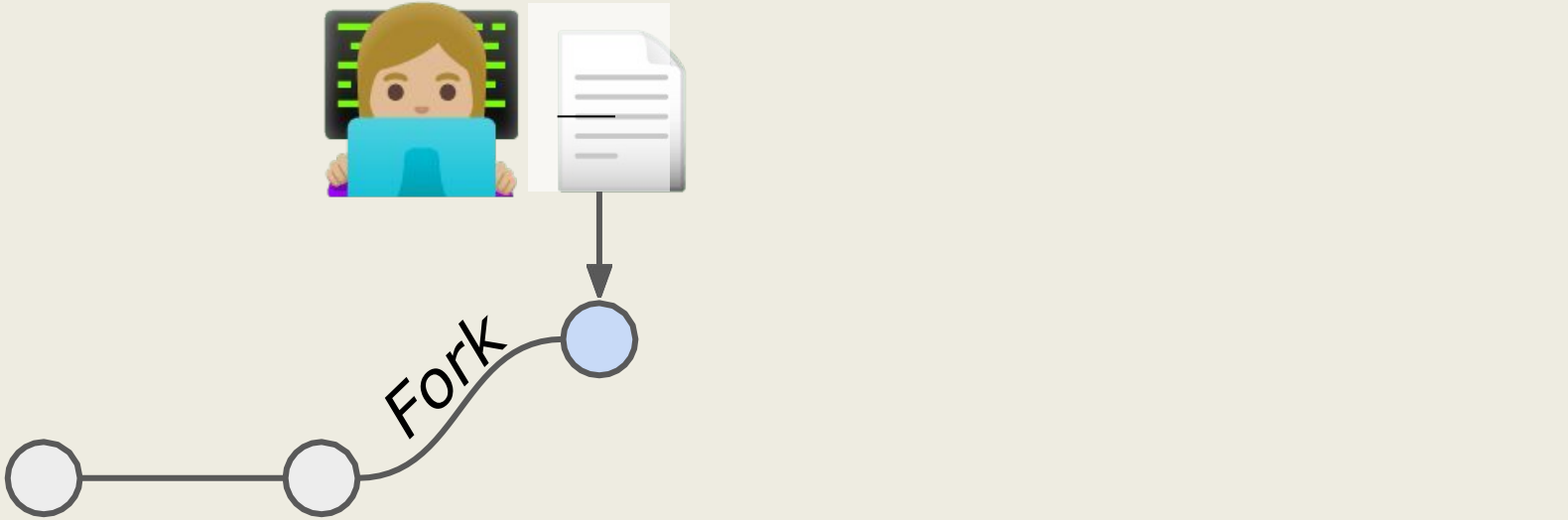


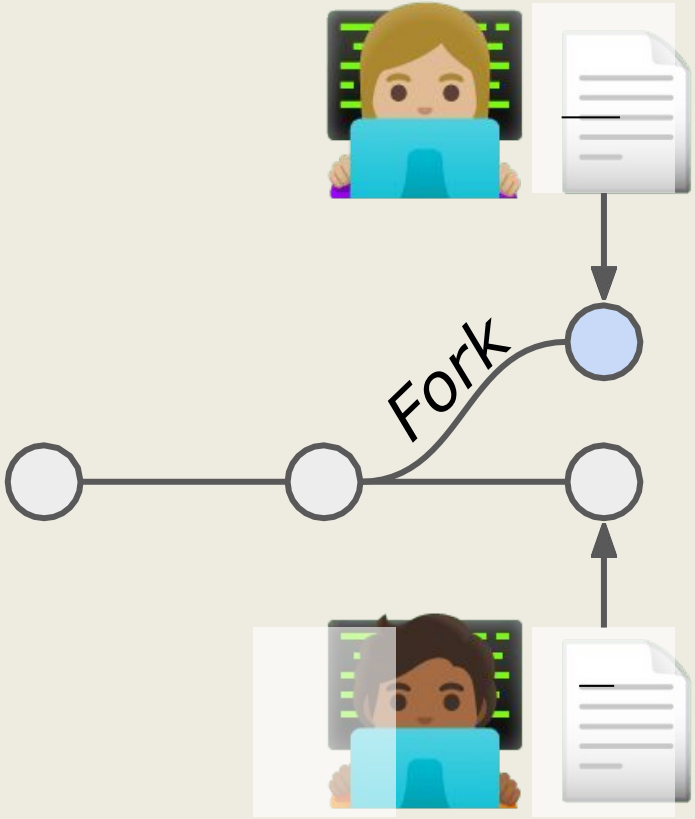
Reference:
https://distrowatch.com/

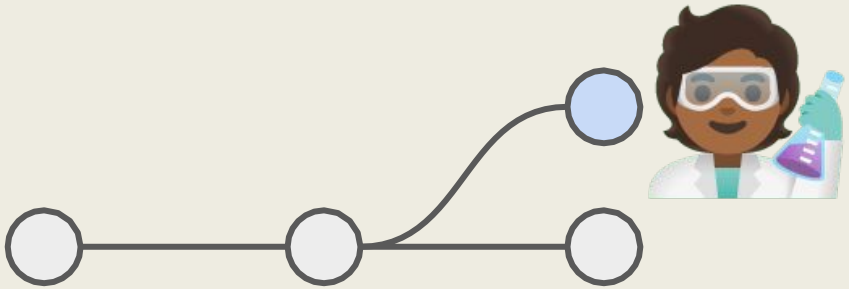

Fork
Merge
Patch

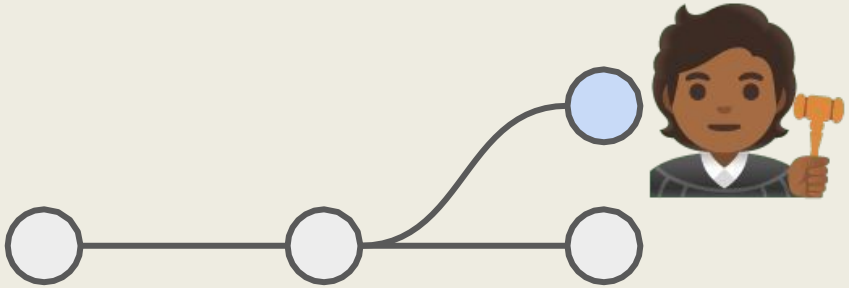


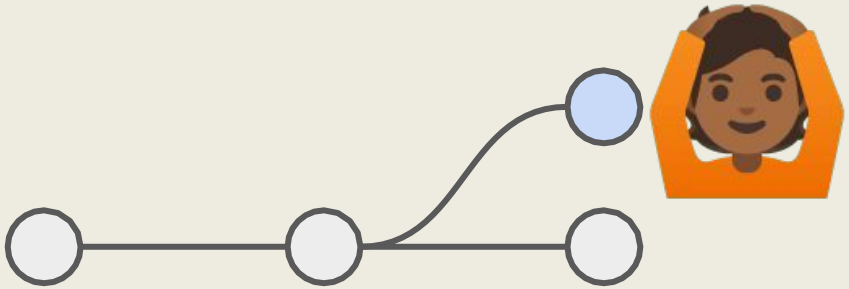

Patc
h

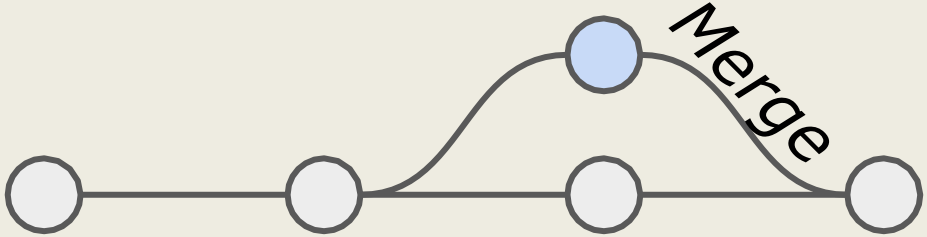

Fork

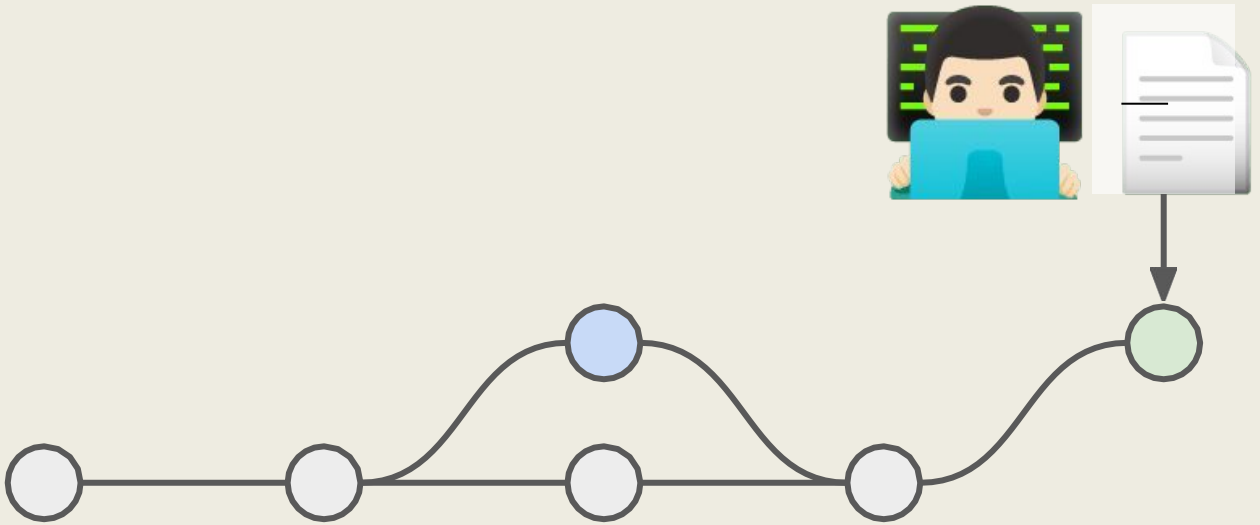

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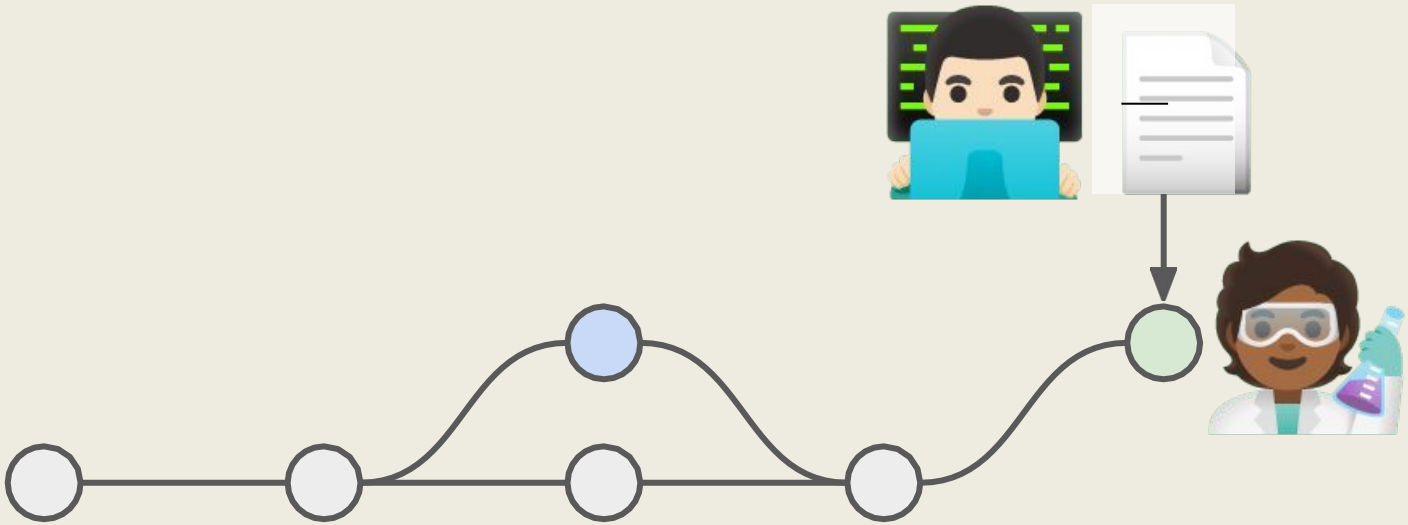


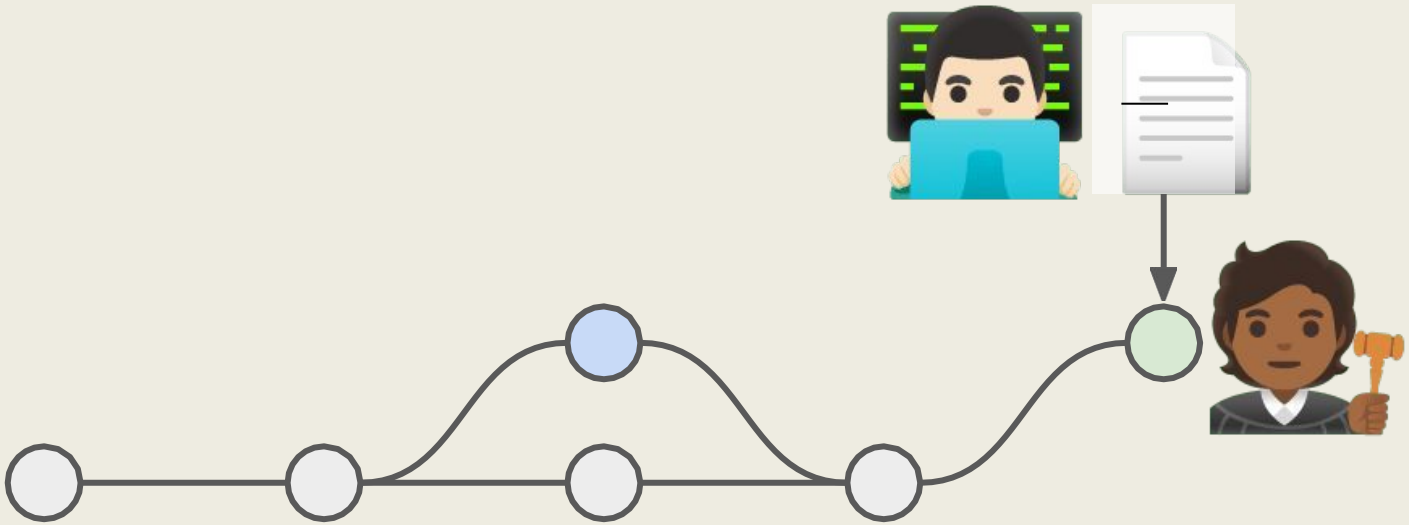


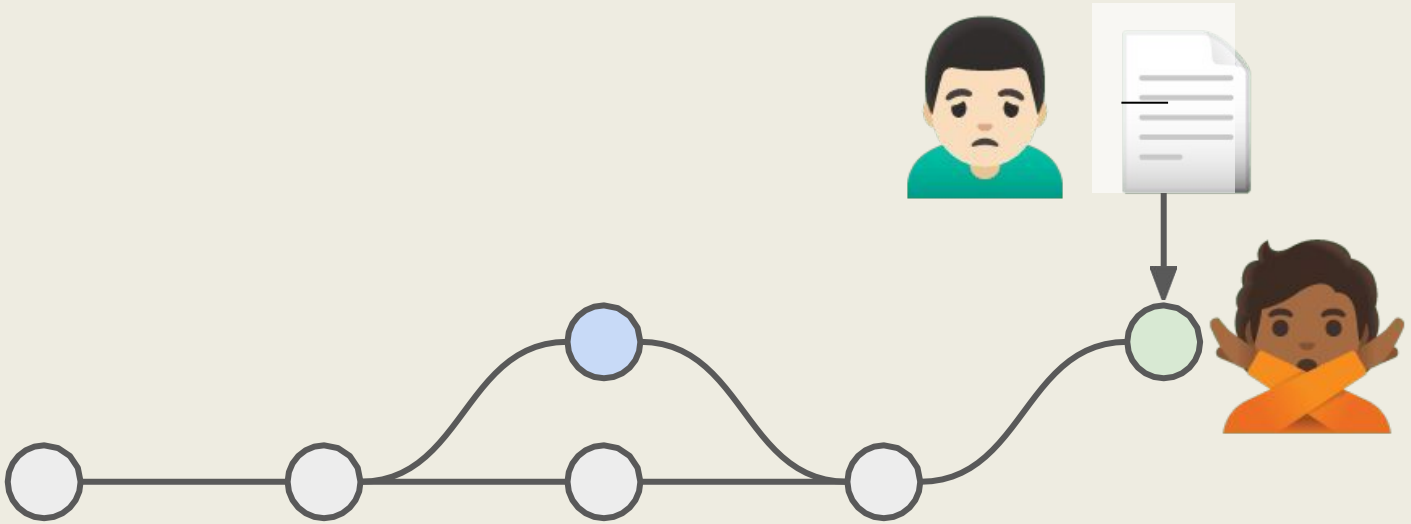


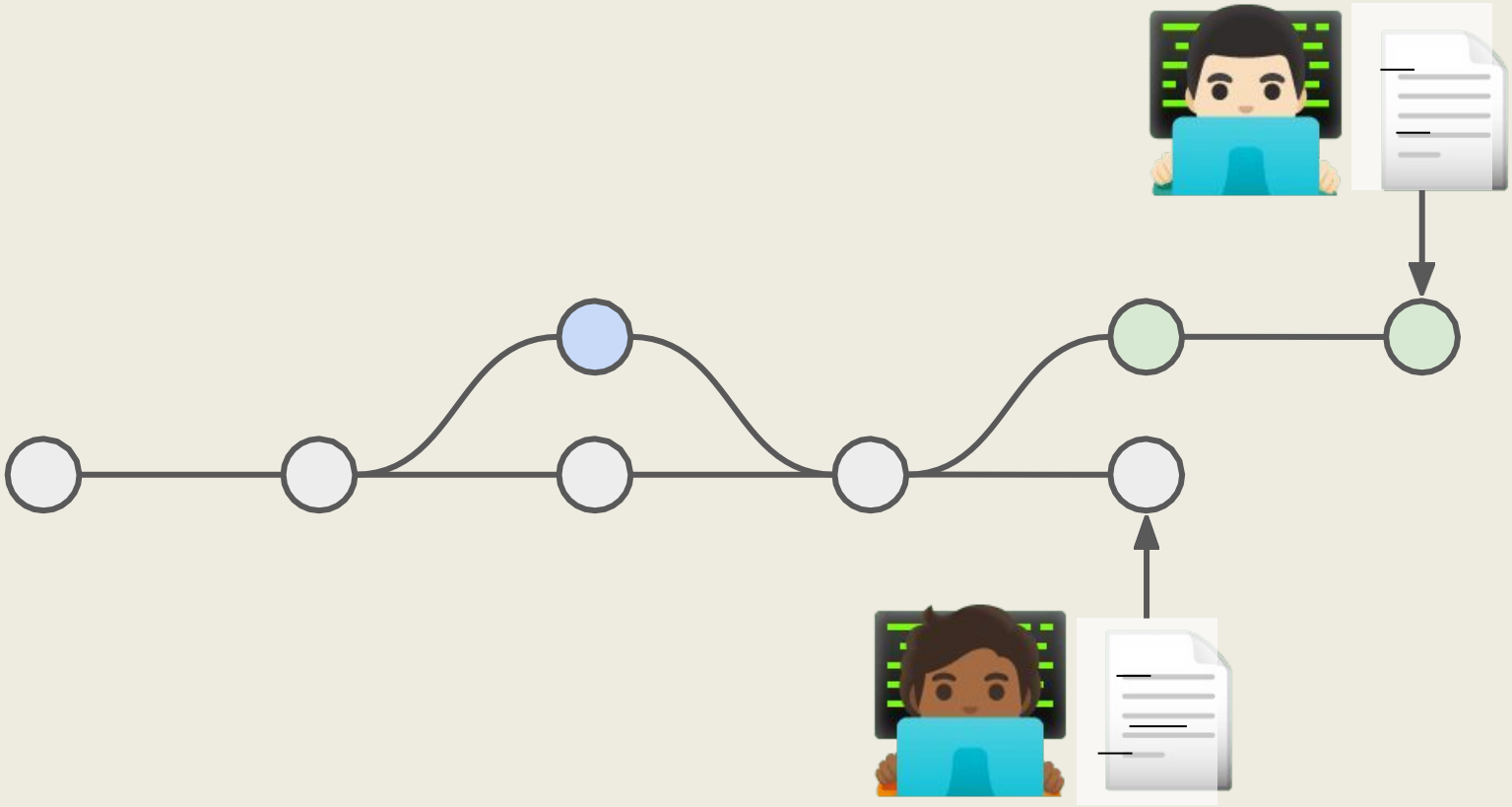

Merge

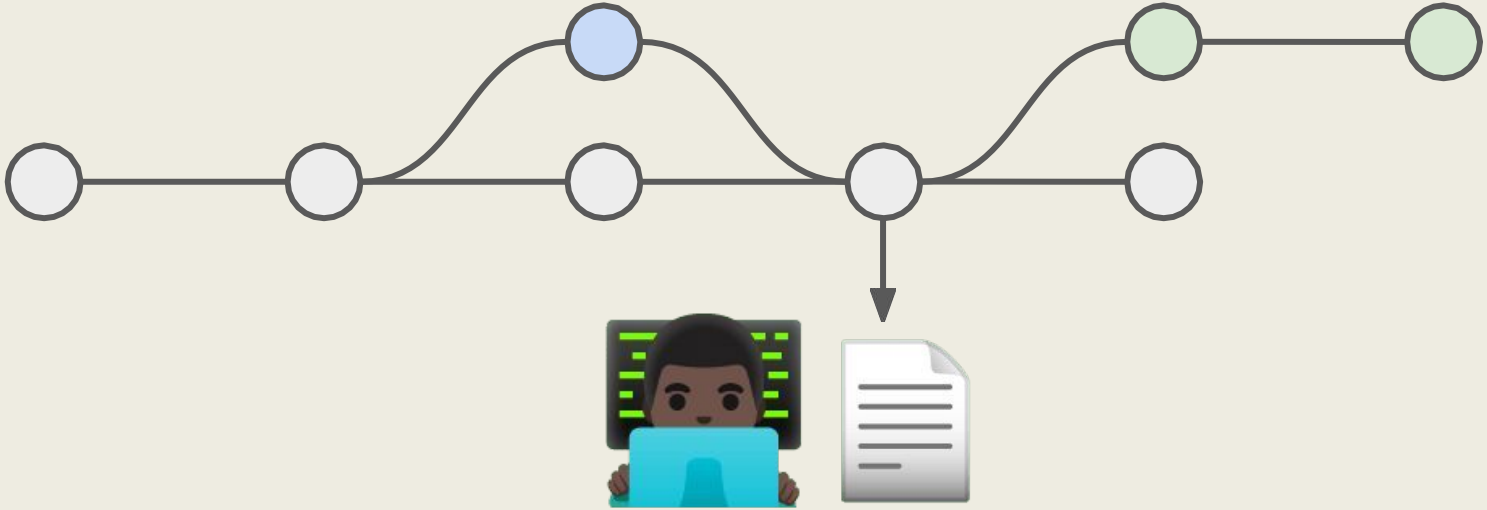













mlflow™

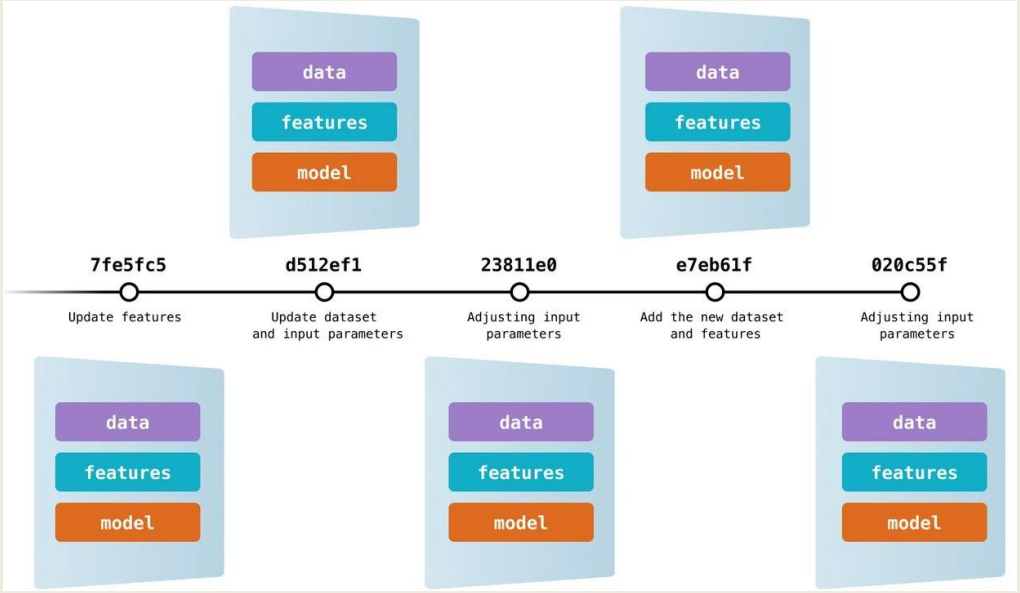

DVC

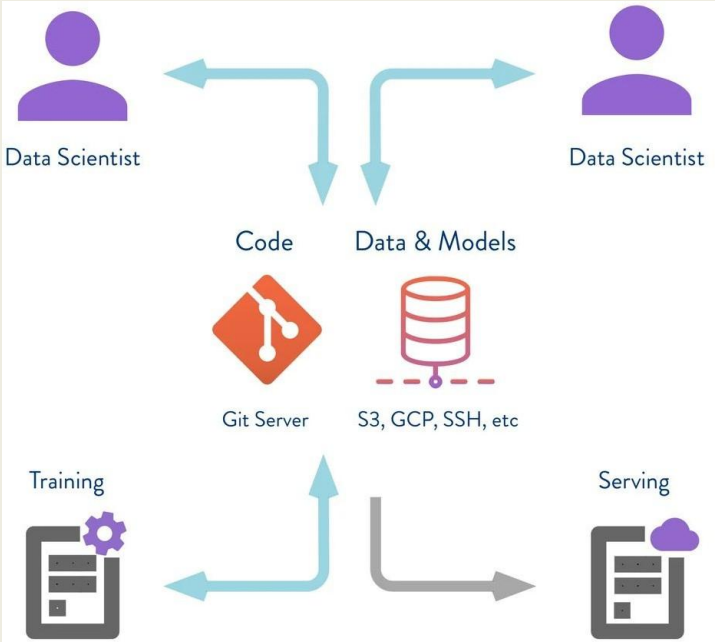

W&B


HUGGING FACE


comet

How DVC's Version Control Works





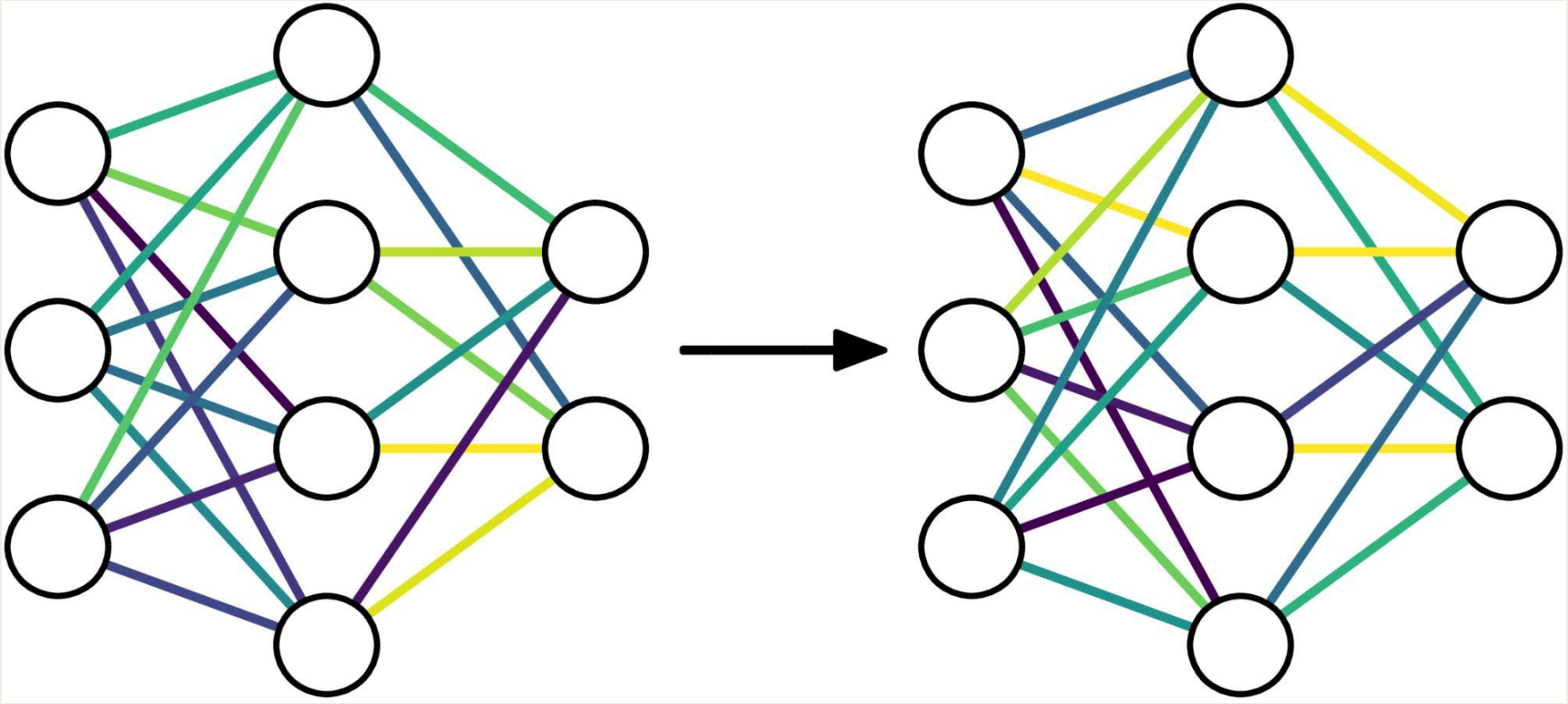
Reference: https://dvc.org/

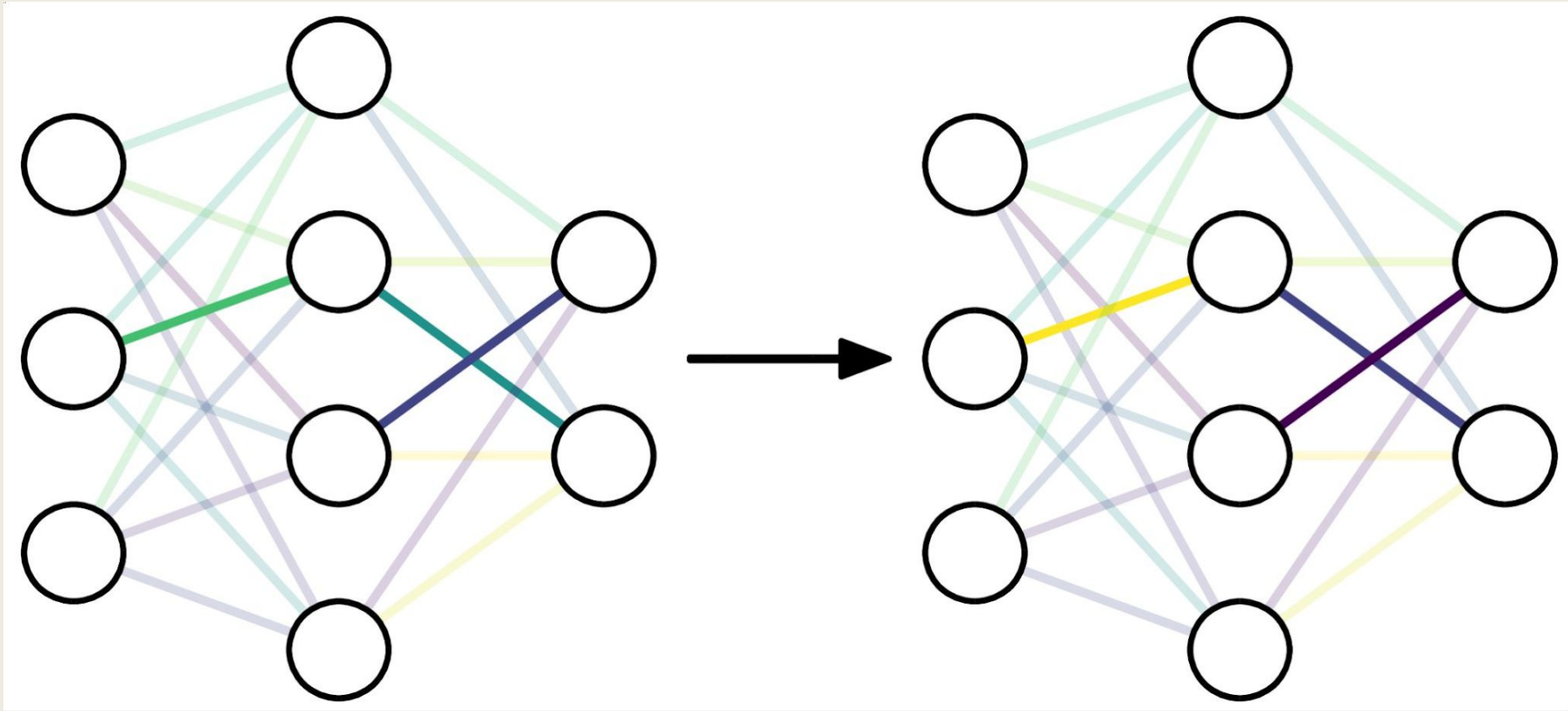
How can we enable collaborative and continual development of machine learning models?

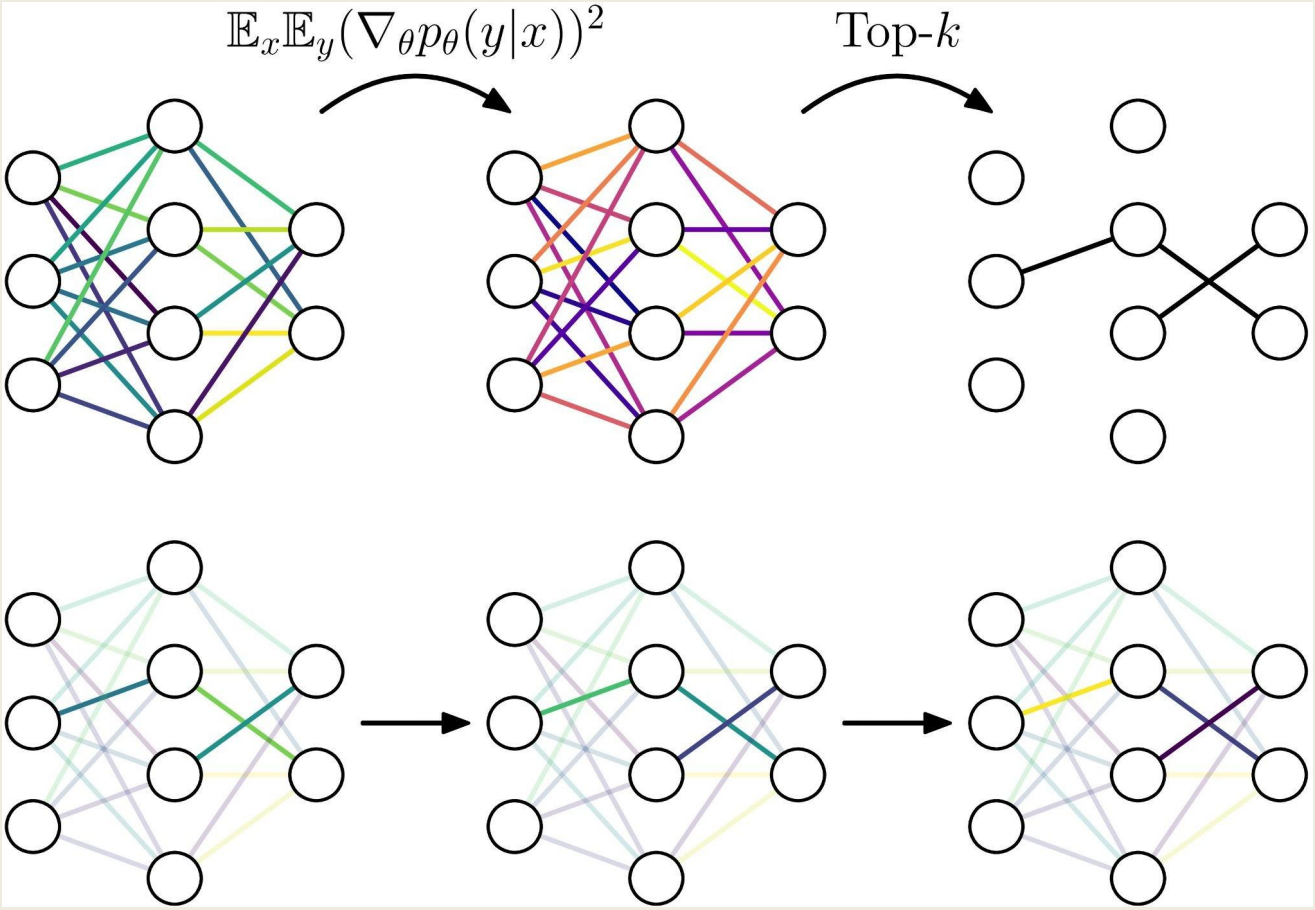
We need to be able to cheaply communicate **patches** and **merge** updates from different contributors.

How can we enable collaborative and continual development of machine learning models?

We need to be able to cheaply communicate **patches** and **merge** updates from different contributors.







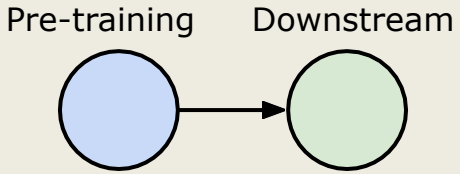
***F**isher-**I**nduced **S**parse Unc**h**anging (FISH) Mask*

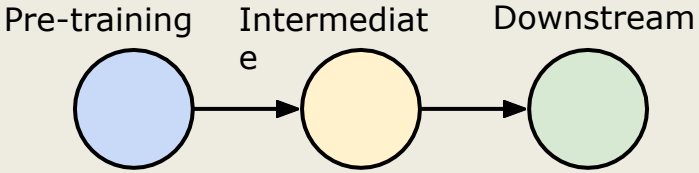
Method	Sparsity	GLUE Score
Dense Fine-tuning	100%	82.5
Bit-Fit	0.08%	81.2
FISH Mask	0.08%	81.3
Diff Pruning	0.50%	81.5
FISH Mask	0.50%	82.6

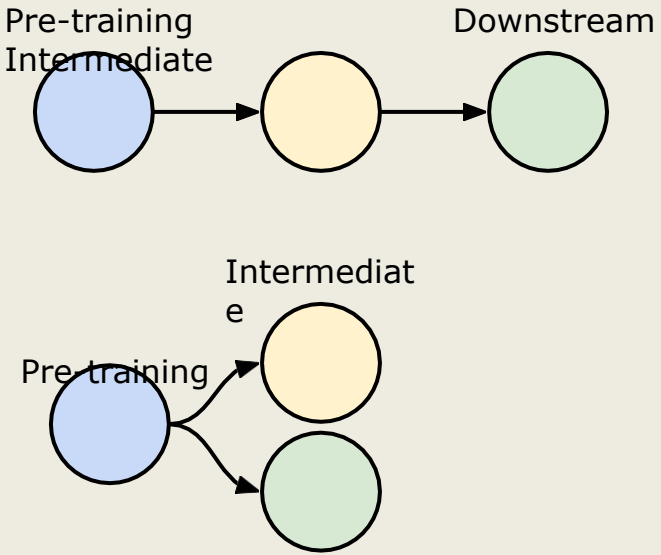
Reference: Training Neural Networks with Fixed Sparse Masks

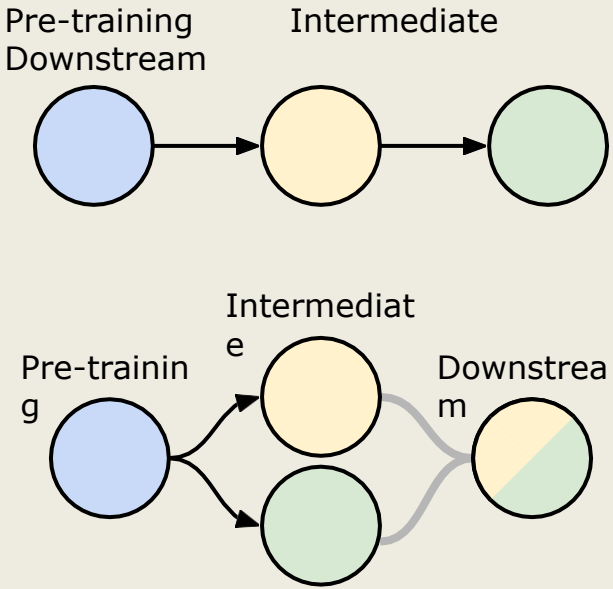
How can we enable collaborative and continual development of machine learning models?

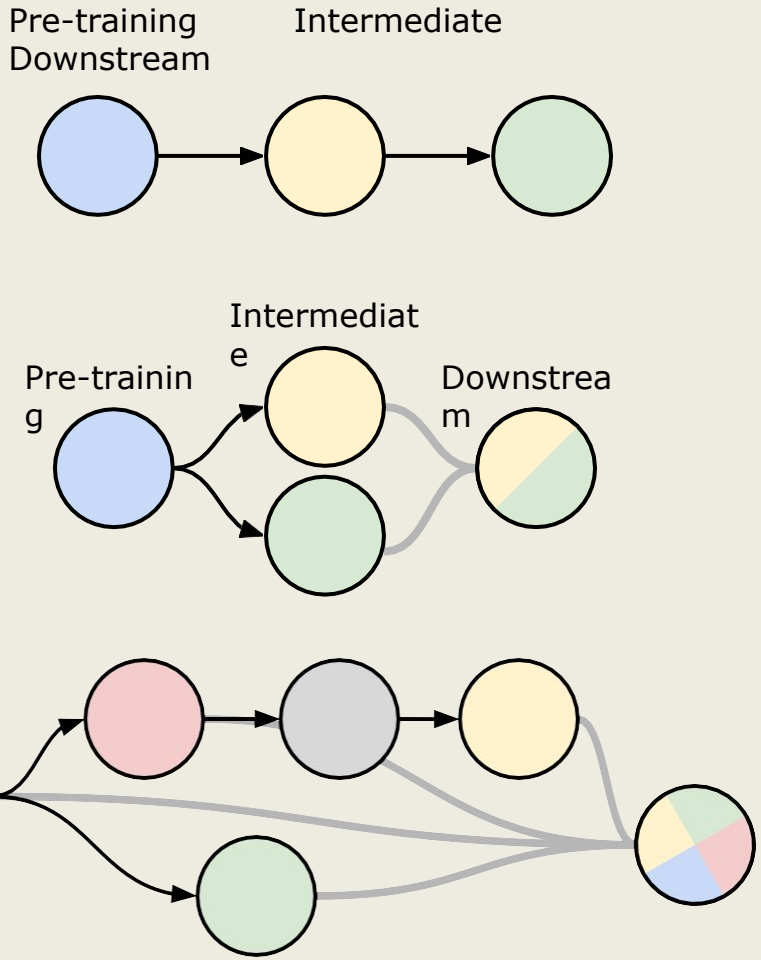
We need to be able to cheaply communicate **patches** and **merge** updates from different contributors.

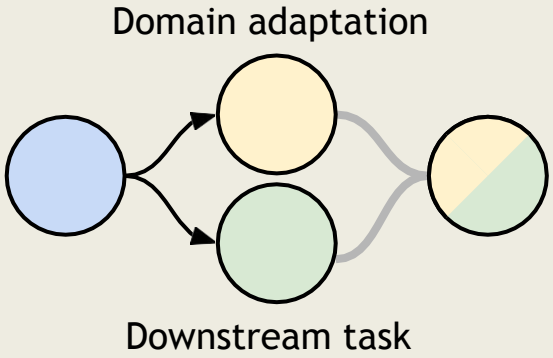

Pre-training
Downstream


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Intermediat
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Downstream


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Pre-training


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Pre-training
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Task	Unmerged	Merged	Fine-tuned
ChemProt	$82.7_{0.3}$	$83.1_{0.4}$	$82.5_{0.1}$
ACL-ARC	$70.5_{3.2}$	$73.2_{1.7}$	$71.5_{3.0}$
SciERC	$81.0_{0.4}$	$81.3_{0.5}$	$81.6_{1.0}$

Reference: *Merging Models with Fisher-Weighted Averaging*

How can we enable collaborative and continual development of machine learning models?

We need to be able to cheaply communicate **patches** and **merge** updates from different contributors.

How can we enable collaborative and continual development of machine learning models?

We need to be able to **rapidly evaluate** proposed changes to the model to ensure backward compatibility.

How can we enable collaborative and continual development of machine learning models?

We need to be able to combine **modular** components of different models to provide new skills and capabilities.

This is a call to ~~action~~ awareness.



Steven Kolawole

Credit to Colin Raffel for the revolutionary vision, original talk, and slides.



Steven Kolawole